PassFrame tuition

Roman Numerals & Negative Numbers

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. CL (C + L = 100 + 50)

2. B — CC + XV = 200 + 15 = 215

3. CCCXL (CCC + XL = 300 + 40)

4. 420 (CD + XX = 400 + 20)

5. B — CD = D − C = 500 − 100 = 400

6. -7°C (12 − 19 = −7)

7. CCLXXV (CC + LXX + V = 200 + 70 + 5)

8. -22°C, -14°C, -3°C, 8°C

9. B, C and E (CCCXLI=341, CDXVI=416 and DXX=520 are greater than 300; CCXCIV=294 and CCLXXX=280 are not)

10. 365 (CCC + LX + V = 300 + 60 + 5)

11. 7°C (−18 + 25 = 7)

12. D — -29°C is the lowest value

13. CDXC (CD + XC = 400 + 90)

14. 644 (D + C + XL + IV = 500 + 100 + 40 + 4)

15. -25°C (−9 − 16 = −25)

Standard

1. DCXXIV (DC + XX + IV = 600 + 20 + 4)

2. 749 (DCC + XL + IX = 700 + 40 + 9)

3. B — CM = M − C = 1,000 − 100 = 900

4. -12°C (15 − 27 = −12)

5. DCCCLXXXVIII (DCCC + LXXX + VIII = 800 + 80 + 8)

6. -40, -24, -3, 11

7. C — CM + XC + IX = 900 + 90 + 9 = 999

8. 5°C (−28 + 33 = 5)

9. CMXCIX (CM + XC + IX = 900 + 90 + 9)

10. 33°C (14 − (−19) = 33)

11. B, C and D (DXLIV=544, DCLXXX=680 and CDXV=415 are all between 400 and 700; CCCXC=390 is below 400 and DCCII=702 is above 700)

12. CDLVI (CD + LVI = 400 + 50 + 6)

13. 897 (DCCC + XC + VII = 800 + 90 + 7)

14. C — 6°C is the highest value

15. -9°C (−17 − 14 = −31; −31 + 22 = −9)

Standard / Hard

1. 999 = CMXCIX; 1 more is 1,000 = M

2. A — M + CD + XL + IV = 1,000 + 400 + 40 + 4 = 1,444

3. 30°C (11 − (−19) = 30)

4. DCCXLIX (DCC + XL + IX = 700 + 40 + 9)

6. A — 400 + 90 + 9 = 499 = CDXCIX

7. CMXCIX (900 + 90 + 9)

8. 19°C, 3°C, -11°C, -36°C

9. A, C and E (DCCXCIX=799, −27°C is colder than −19°C, and CDXLIV=444 are true; MCDXCII is actually 1,492 (not 1,392) and −8°C is colder than 5°C, not warmer)

11. Floor -2 (4 − 11 = −7; −7 + 5 = −2)

12. C — −22 + 13 = −9; −9 + 15 = 6

13. 27 BC = −27; from −27 to 14 is 27 + 14 = 41 years

14. 645 (DCCCXLV = 845; 845 − 200 = 645)

5. -104 m (−85 + 32 = −53; −53 − 51 = −104)

10. 1,994 (M + CM + XC + IV = 1,000 + 900 + 90 + 4)

15. 13°C (−34 + 47 = 13)

Homework — Answers

Easy / Standard

1. CXXV (C + XX + V = 100 + 20 + 5)

2. C — C + LX + VI = 100 + 60 + 6 = 166

3. CCXVIII (CC + X + VIII = 200 + 10 + 8)

4. 344 (CCC + XL + IV = 300 + 40 + 4)

5. B — D represents 500

6. -6°C (9 − 15 = −6)

7. CCLX (CC + LX = 200 + 60)

8. -20°C, -12°C, -1°C, 6°C

9. B, C and E (CCLXXX=280, CCCXV=315 and CDX=410 are greater than 250; CCXLI=241 and CCXIX=219 are not)

10. 372 (CCC + LXX + II = 300 + 70 + 2)

11. 6°C (−16 + 22 = 6)

12. D — -25°C is the lowest value

13. CDXXX (CD + XXX = 400 + 30)

14. 496 (CD + XC + VI = 400 + 90 + 6)

15. -22°C (−8 − 14 = −22)

Standard

1. DLV (D + L + V = 500 + 50 + 5)

2. 693 (DC + XC + III = 600 + 90 + 3)

3. A — CM + L = 900 + 50 = 950

4. -12°C (17 − 29 = −12)

5. DCCLXXVII (DCC + LXX + VII = 700 + 70 + 7)

6. -48, -33, -6, 9

7. B — CM + XL + IV = 900 + 40 + 4 = 944

8. 9°C (−31 + 40 = 9)

9. CMIX (CM + IX = 900 + 9)

10. 37°C (16 − (−21) = 37)

11. B, C and E (DXCIX=599, DCCLIII=753 and DCCLXXXV=785 are all between 500 and 800; CDLX=460 and CCCXCIX=399 are both below 500)

12. DCXVIII (DC + X + VIII = 600 + 10 + 8)

13. 912 (CM + XII = 900 + 12)

14. C — 7°C is the highest value

15. -11°C (−19 − 17 = −36; −36 + 25 = −11)

Standard / Advanced

1. 576 (CDLXXVI = 476; 476 + 100 = 576)

2. B — 5 − 18 = −13°C

3. 623 = DCXXIII; 150 less is 473 = CDLXXIII

4. 434 (CCXVII = 217; 217 × 2 = 434)

5. C — −26 + 39 = 13°C

6. 344 = CCCXLIV; 344 + 155 = 499 = CDXCIX

7. 40°C (17 − (−23) = 40)

8. A, B, C and D (DCCXXXI=731; MCXLIV=1,144; −29°C is 12°C colder than −17°C (−17−12=−29); 6−(−27)=33 so 6°C is 33°C warmer than −27°C — all correct; but CDXCVI actually represents 496, not 486)

9. DCCXLII = 742; 742 − 175 = 567 = DLXVII

10. A — −58 + 21 = −37; −37 − 39 = −76

11. 683 = DCLXXXIII; 683 − 256 = 427 = CDXXVII

12. 14°C to -22°C is a drop of 36°C; half of 36 is 18; -22 + 18 = -4°C

13. B — DCCXCIX = 799; 799 × 2 = 1,598

14. £75 (−184 + 259 = 75)

15. 856 = DCCCLVI, 349 = CCCXLIX; difference = 856 − 349 = 507 = DVII